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Introduction to networks

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Network devices

Who connects to the network?

Network devices

- Clients (workstations) / Servers
- Network devices

koncetrator



Hub-PT
Hub0

przełącznik



2960-24TT
Switch0

router



1941
Router0



Laptop-PT
Laptop0



Server-PT
Server0

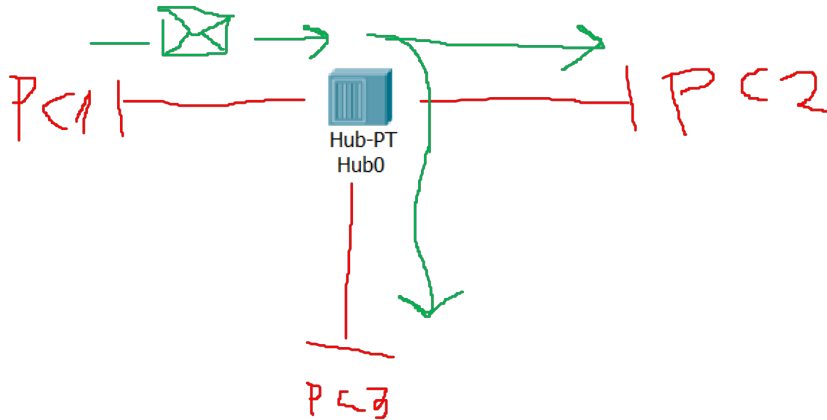


PC-PT
PC0

Hosts - network devices with an IP address (routers, workstations)

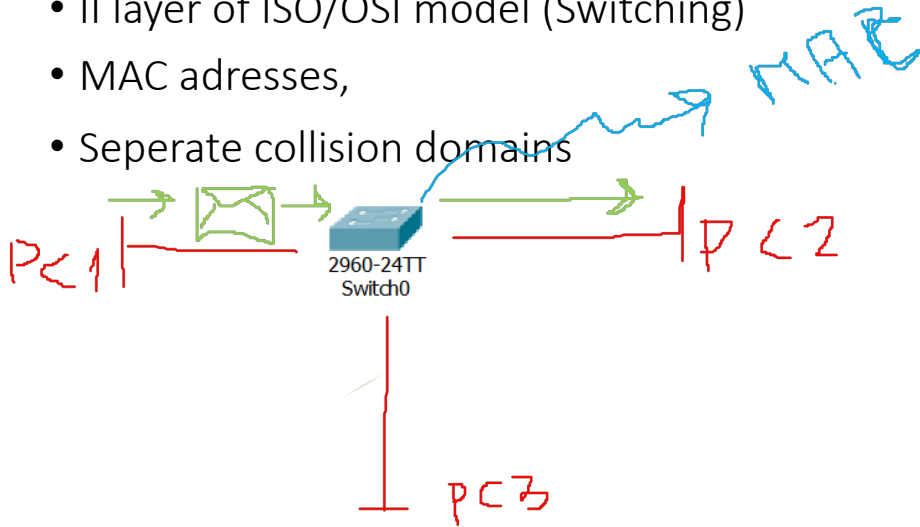
Koncentrator / Hub

- 1 layer of ISO/OSI model
- Hubs form collision domains!!



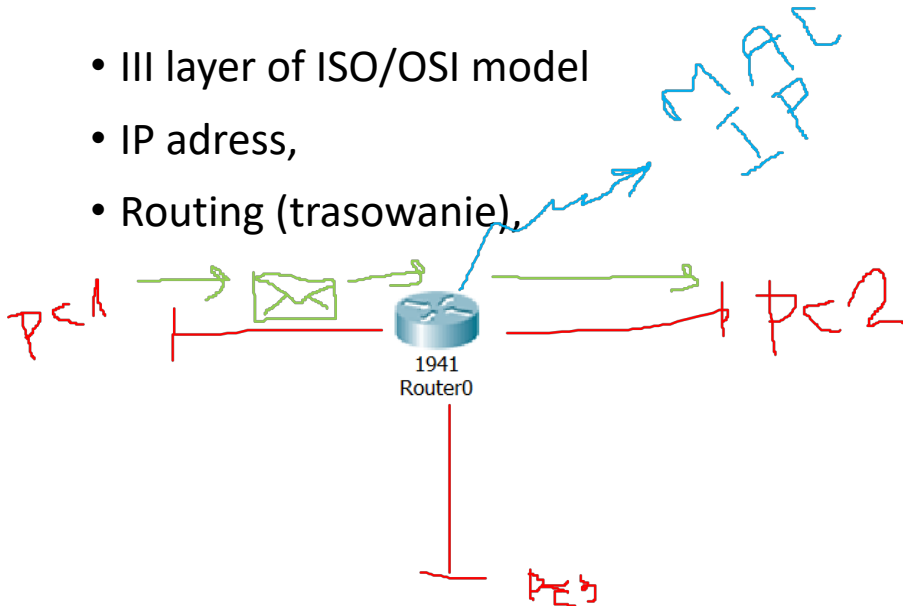
Przełącznik / Switch

- II layer of ISO/OSI model (Switching)
- MAC addresses,
- Seperate collision domains

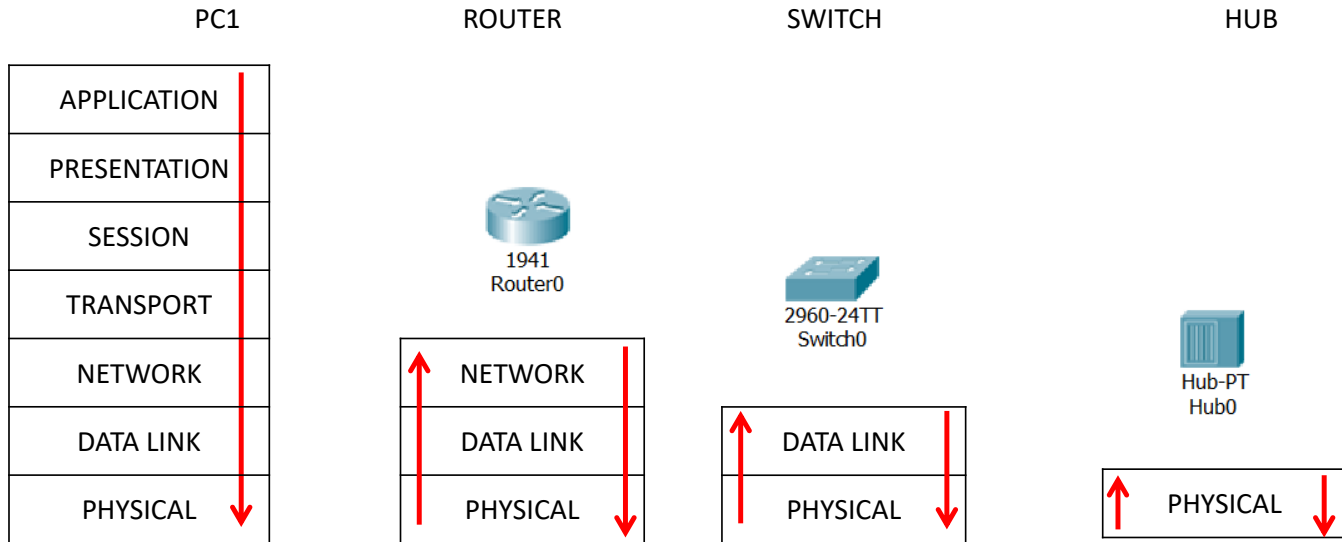


Router

- III layer of ISO/OSI model
- IP address,
- Routing (trasowanie),



Network devices and ISO/OSI model

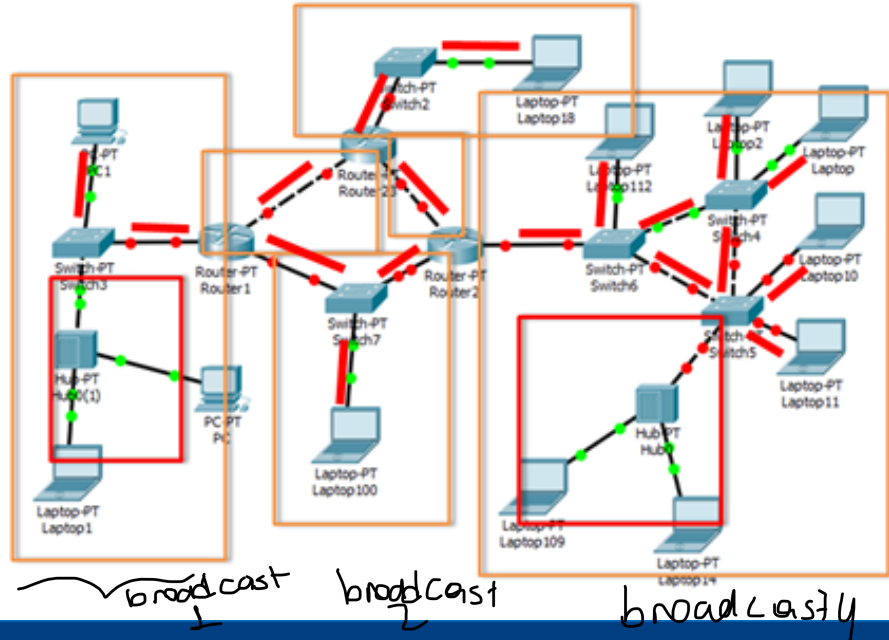


Broadcast domains vs collision domains

Broadcast domains
(domeny rozgłoszeniowe)

Collision domains
(domeny kolizji)

każdy switch to collision domain
każdy router to collision domain
Hub'a bağlı tüm cihazlar
1 collision domain



Network devices in practise...



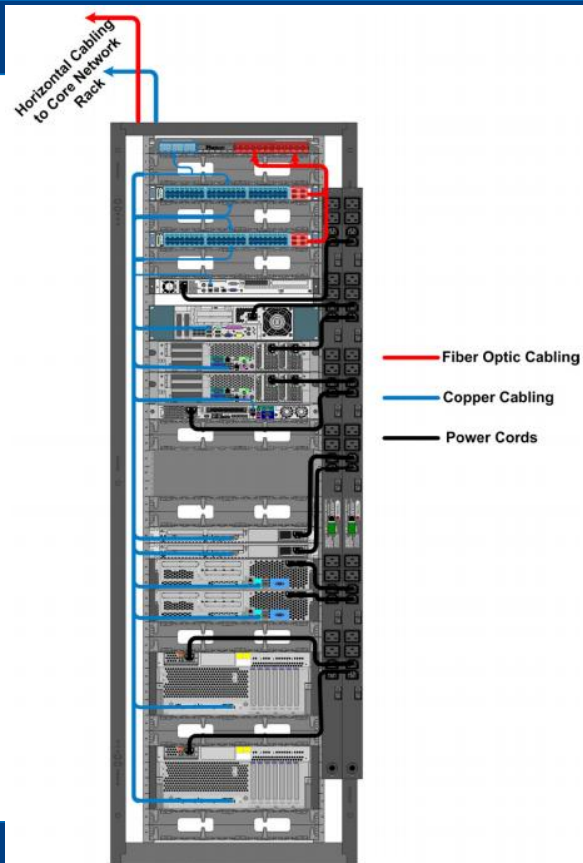
Devices working as routers used in home or SOHO (Small Office / Home Office) they combine a router, switch, access point, firewall ...



The networks supporting companies and institutions are built in network cabinets and consist of devices dedicated to each task separately.

Wiring the cabinets CISCO

An example of a CISCO cabinet with cabling >>>



Inside CISCO network devices

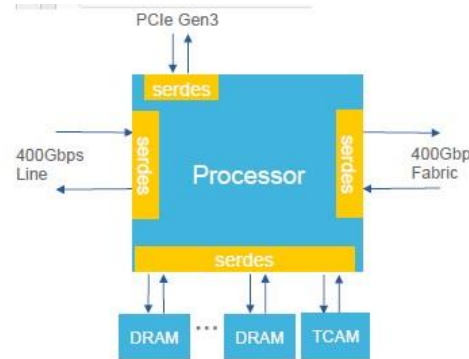
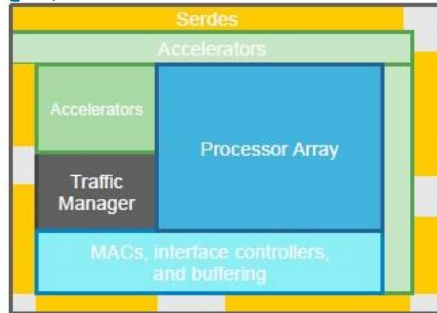
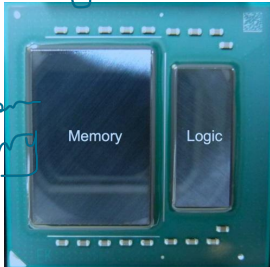
Physical construction of devices CISCO

Router contains:

- **Processor**, → Kontrol düzlemi (yönetim, routing tablosu)
- **ASIC**, → veri düzlemi (paketlerin hızlı ve verimli iletimi)

Memories:

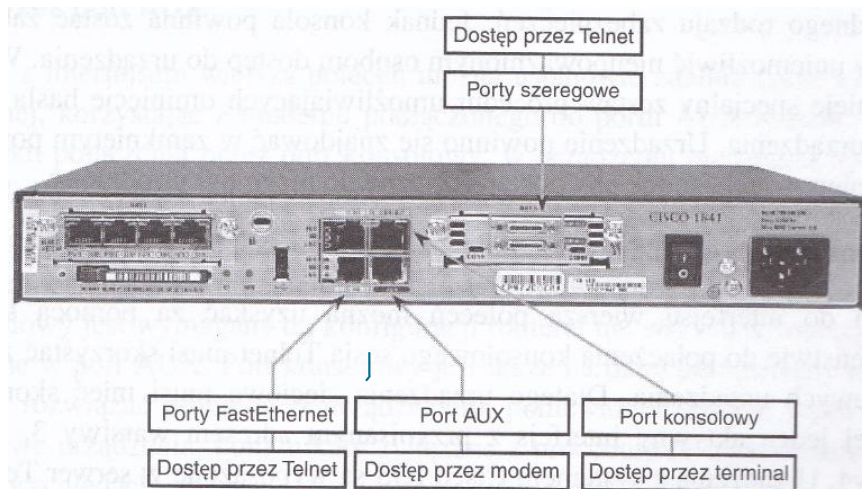
- ROM, → read-only memory
- flash,
- RAM, → random access memory
- NVRAM, → non-volatile random access memory



Communication methods in IOS

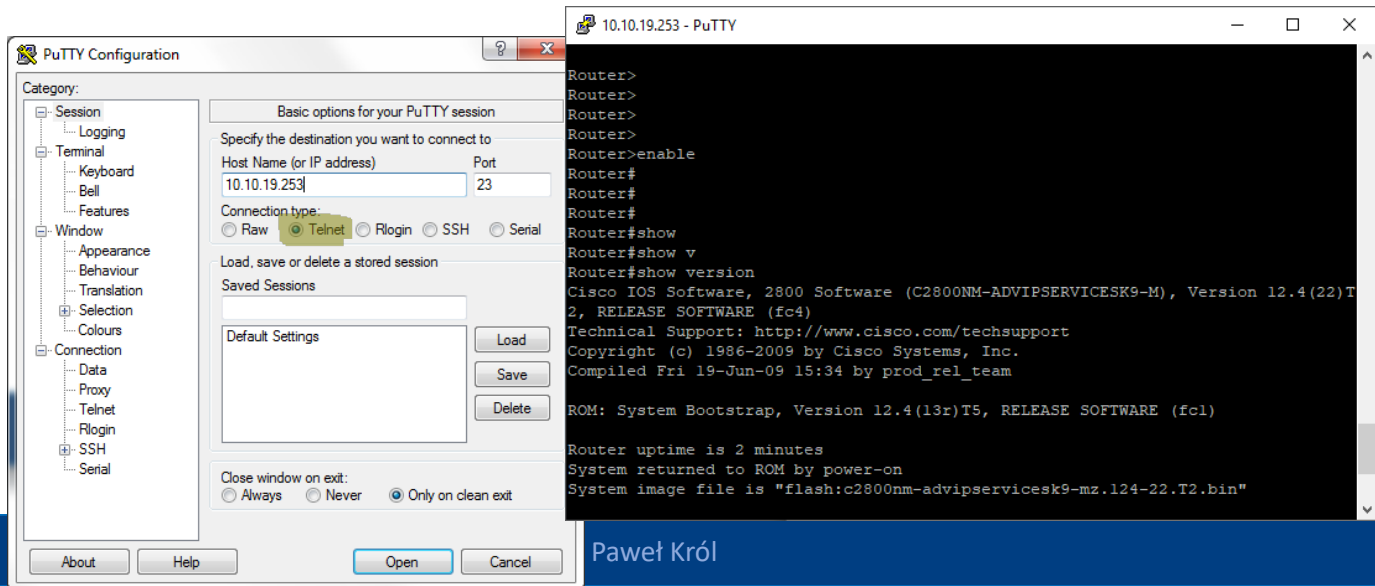
Ways of gaining access to CISCO network devices:

- Access via the console port
- Access via Telnet or SSH
- Access via the AUX port
- Access via USB
- Other...



Telnet/SSH connection

We will use the PUTTY program to connect the devices via telnet protocol



System IOS – Internetwork Operating System

IOS Internetwork Operating System

The operating system used in the CISCO devices

Basic IOS commands

Basic IOS commands

Router> ! comment

Router> ?

Router> show ?

Basic IOS commands

Router> show clock

Router> show version

Router> show history

Router> show users

Router> show running-config

Router> show startup-config

Router> show interfaces

Router> show ip interface brief

RAM'de bulunur. belirtir
cihazın şu anda
çalışan (aktif) Konfigürasyonunu
gösterir

Cihaz yeniden açıldığında
yükleyeceği yapılandırma'yı
gösterir. Non-volatile (NVRAM) üzerinde
bulunur

Users' permissions

Router> ! user EXEC level

Router# ! privileged EXEC level

Router(config)# ! global configuration level

Router(config-if)# ! specific configuration level

Verification of the configured interface

```
Router>  
Router> enable  
Router#  
Router# configure terminal  
Router(config)# ?  
Router(config)# interface FastEthernet 0/0  
Router(config-if)# ?  
Router(config-if)#  
Router(config-if)# end  
Router# ?  
Router#  
Router# disable  
Router> ?  
Router> exit
```

The diagram illustrates the sequence of commands and mode transitions in a Cisco router configuration. Green arrows point from the command lines to descriptive text boxes:

- From `enable` to `Enable privileged mode`
- From `configure terminal` to `Enable configuration mode`
- From `interface FastEthernet 0/0` to `Interface settings FastEthernet 0/0`
- From `end` to `Exit from the configuration mode`
- From `disable` to `Exit from privileged mode`

Switch – analysis of MAC table

```
Switch> show mac address-table
```

```
Switch# clear mac address-table
```



Configuration of interfaces in router

```
Router>
Router> enable
Router#
Router# configure terminal
Router(config)#
Router(config)# interface FastEthernet 0/0
Router(config-if)# ip address 192.168.0.1 255.255.255.0
Router(config-if)#
Router(config-if)# no shutdown
Router(config-if)# end
Router# show ip interface brief
Router# disable
Router>
Router> exit
```



IP address and netmask

Switching on the interface

Verification of the configured interface



Programming of switch virtual interfaces (SVI)

```
Switch>
Switch> enable
Switch#
Switch# configure terminal
Switch(config)#
Switch(config)# interface vlan1
Switch(config-if)# ip address 192.168.0.1 255.255.255.0
Switch(config-if)#
Switch(config-if)# no shutdown
Switch(config-if)# end
Switch# do sh ip int br
Switch# disable
Switch>
Switch> exit
```

przełącznik



2960-24TT
Switch0

Router – loopback configuration

```
Router# conf t
```

```
Router(config)# interface Loopback 0
```

```
Router(config-if)# ip address 192.168.0.1 255.255.255.255
```

```
Router(config-if)# end
```

```
Router# show ip int brief
```

```
Router# show running-config
```

```
Router# show run
```

Do not use command „no shutdown”

```
!
interface Loopback1
 ip address 172.16.1.1 255.255.255.128
!
interface Loopback2
 ip address 172.16.1.129 255.255.255.128
!
interface FastEthernet0/0
 ip address 172.16.0.2 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 10.10.0.2 255.255.255.252
 duplex auto
 speed auto
--More-- |
```

Example project in Packet Tracer

